



# Uliana Hrab

DATA · MACHINE LEARNING · C++

I am in my fourth year of a BSc in Applied Mathematics, based in Norway. The degree covered a wide range of subjects: machine learning and deep learning, computer vision, blockchain, computational geometry, C++ and some economics. Data work is what I like most, so I am looking mainly for analyst and ML/DL roles, though I am open to other positions. I am ready to learn on the job and to take additional courses for a qualification, with support from NAV.

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## EMAIL

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## PHONE

+47 46 55 57 59

## LOCATION

Norway

## GITHUB

github.com/draurug

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## SELECTED WORK

| Longer write-ups of each project are on the work page.

### Ethereum Fraud Detection

2026

Classified 9,841 Ethereum addresses as fraudulent or legitimate from 50 transaction-derived features. LightGBM reached F1 0.987 on held-out data. Before that, I found and fixed a leak: a naive random split put 5.7% of test rows into the training set through duplicate feature profiles. I made the split group-aware and selected models by cross-validated PR-AUC instead of test score.

Python · scikit-learn · LightGBM · XGBoost · pytest

### CrossW

2026

Designed and shipped a themed crossword site, from the product idea to a live deployment. Four content packs, 75 unit tests, roughly 5,400 lines of TypeScript. Answers are compiled to opaque indices at build time, so they never reach the browser in readable form.

Next.js 15 · React 19 · TypeScript · Vitest · Cloudflare Workers

### Voronoi Path Planner

2026

Interactive path planner that optimises for clearance instead of distance. It builds a clearance field against the obstacles and binary-searches the largest threshold that keeps start and goal connected. Dijkstra then runs inside that corridor. Comes with a generated figure set and a written report.

C++ · OpenGL · FreeGLUT · NumPy · SciPy

### Networked Tic-Tac-Toe and Battleship

2026

Two Qt desktop games in C++. The first is a client for a multiplayer game server: invitation handshake, turn synchronisation and game-over detection, on top of a third-party Boost.Asio transport from the course. In the second, the game rules are entirely free of Qt and the widget only paints.

C++17/20 · Qt · CMake

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## TECHNICAL SKILLS

### LANGUAGES

Python, C++ (17/20), TypeScript, JavaScript, SQL, C#, Octave/  
MATLAB, LaTeX

### WEB AND TOOLING

Next.js, React, Vue, Node, PostgreSQL, Git, CMake, pytest,  
Vitest, Docker, Cloudflare Workers

### DATA AND MACHINE LEARNING

scikit-learn, LightGBM, XGBoost, pandas, NumPy, SciPy,  
matplotlib, seaborn, Jupyter, Power BI

### OTHER

OpenGL, Qt, Boost.Asio, Unity and C#, Blender, Arduino

## EDUCATION

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### BSc Applied Mathematics — Odesa I. I. Mechnikov National University

2023 — present

Currently in the fourth year. Institute for Optimal Control and Economic Cybernetics. Coursework in numerical methods, computational geometry and computer graphics, decision theory, blockchain technology and machine learning. My course paper was on fraud detection. I also did a bonus project on Voronoi-based path planning.

### IT Step Academy — IT-focused secondary programme

2019 — 2022

Programming fundamentals, game development in Unity and C#, 3D modelling in Blender, and Arduino.

### Odesa Lyceum no. 1 (A. P. Bystrina) — secondary education

2016 — 2023

English specialisation.

## EXPERIENCE

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### Logistics manager — SmartAuto LLC

May — Aug 2024

Coordinated vehicle logistics. Maintained the operational records the scheduling ran on.

### Mentor — IT2School

Dec 2023 — Jan 2025

Taught programming to school students and helped them through their assignments.

### Project lead — Mechnikov student card

2023 — present

Led the student card project at the university. Member of the Student Union and of AtomSpace.

## LANGUAGES

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- **Russian** and **Ukrainian** — native
- **English** — C1, certified by the London School of English (2025)
- **Norwegian** — B2 course completed, Læringssenteret for voksne i Ringerike kommune; Norskprøven passed
- **German** — Goethe-Institut certificate

## CERTIFICATES

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- English C1 — London School of English, 2025
- Working with Data in Python — DataCamp, 2024
- Data Visualisation in Power BI — DataCamp, 2024
- Data Analysis in Google Sheets — DataCamp, 2024
- C++ Intermediate — Sololearn, 2023
- C++ Basics — Coursera, 2023

## REFERENCES

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Available on request.